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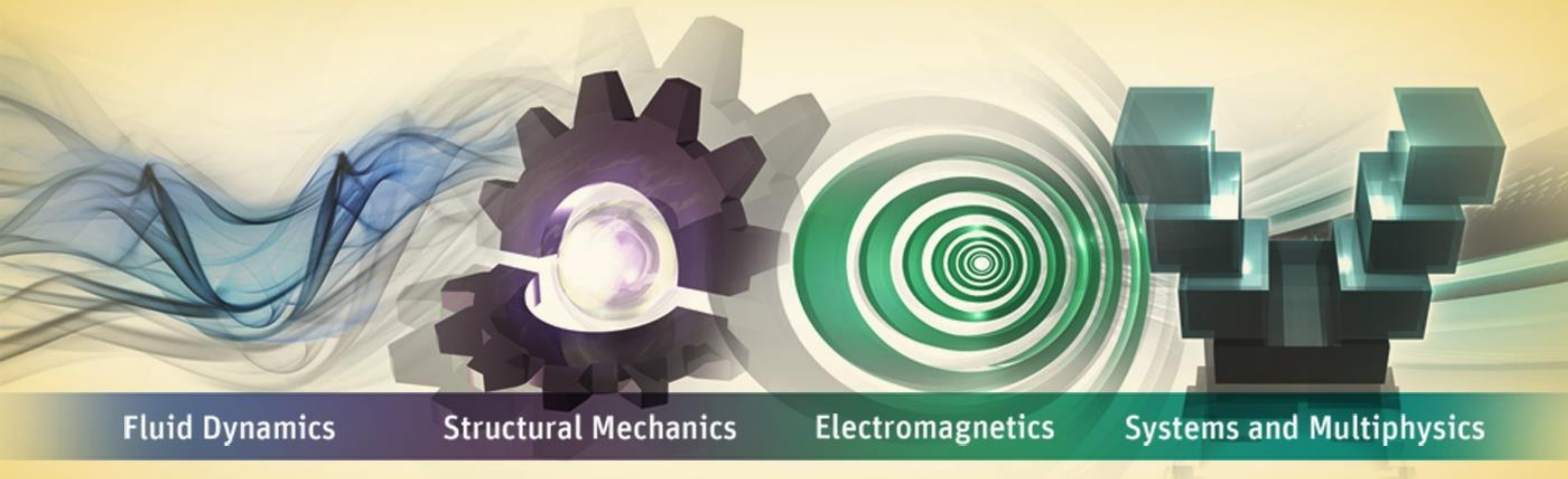
Workshop 3B: Symmetric vs. Asymmetric



ANSYS Mechanical
Introduction to Structural Nonlinearities

Workshop 3B: *Symmetric vs. Asymmetric*

Goal

Use contact to predict pressure profile at spherical interface between ball and socket.

Compare and contrast Symmetric vs. Asymmetric Contact behavior.

Model Description

2D Axisymmetric model of ball and socket joint

Materials:

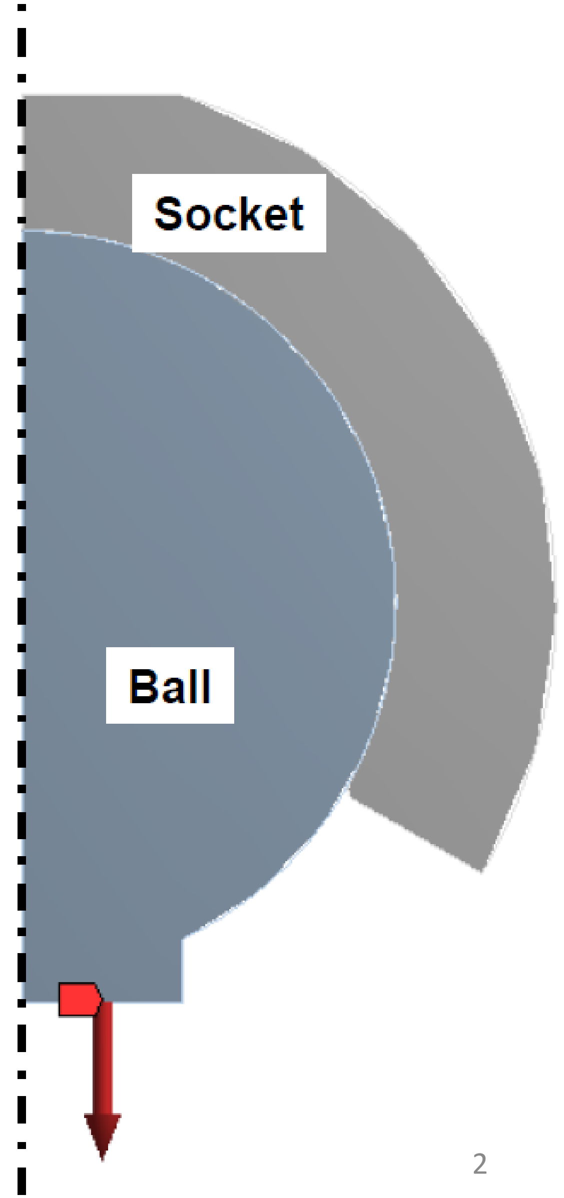
- Steel on Steel

Contact between parts:

- One frictional region on spherical interface
- 0.40 coefficient of friction

Loads & Boundary Conditions:

- Socket is fixed at the top
- Ball has a 1000N force applied to it in vertical (-Y) direction



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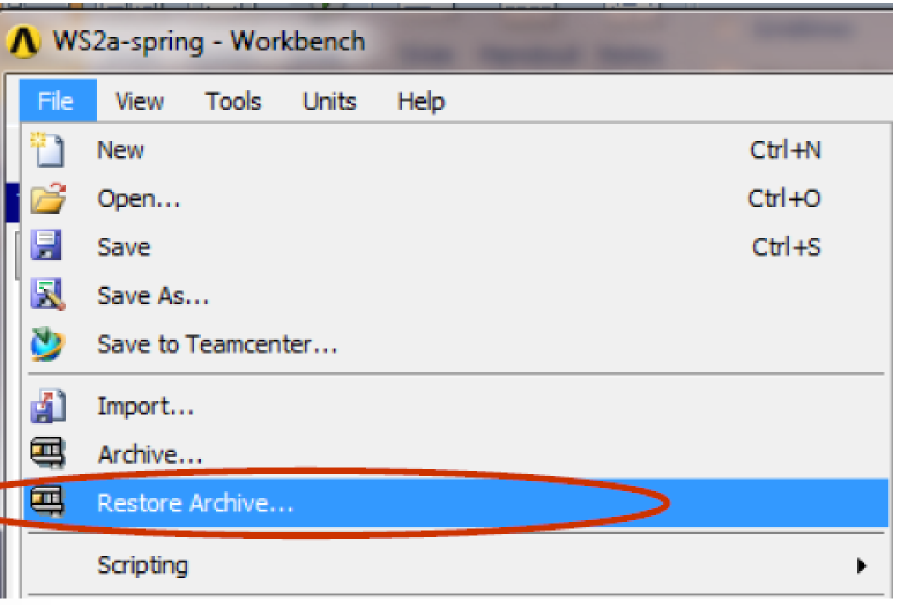
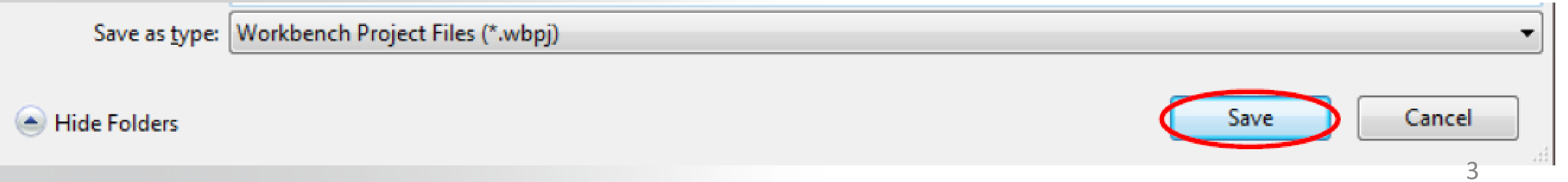
Steps to Follow:

Restore Archive... browse for file “SNL WS3b-Socket.wbpz”



Save as

- **File name: “WS3b-Socket”**
- **Save as type: Workbench Project Files (*.wbpj)**



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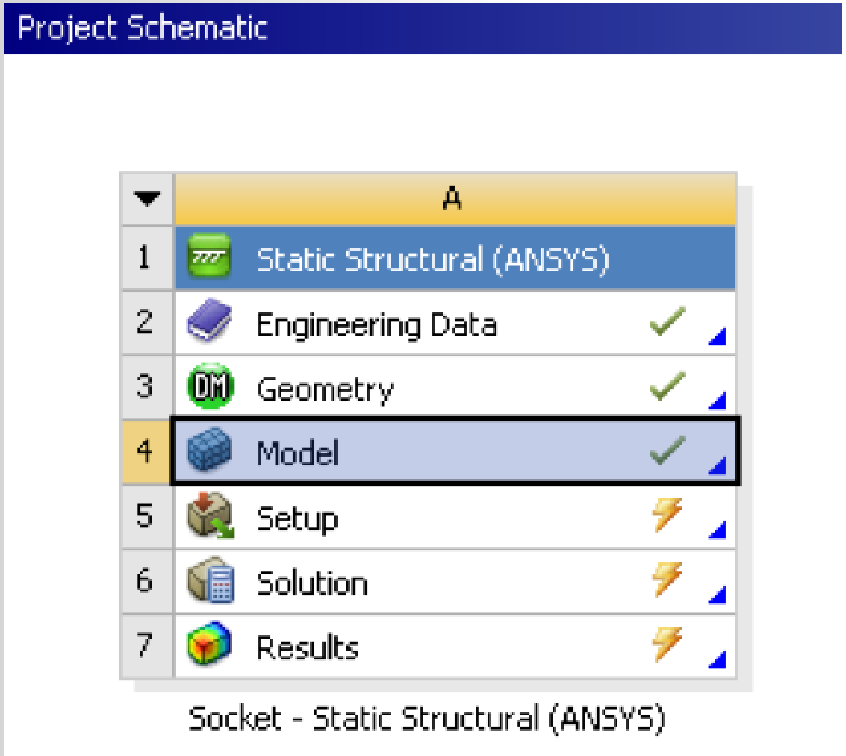
The Project Schematic should look like the picture below.

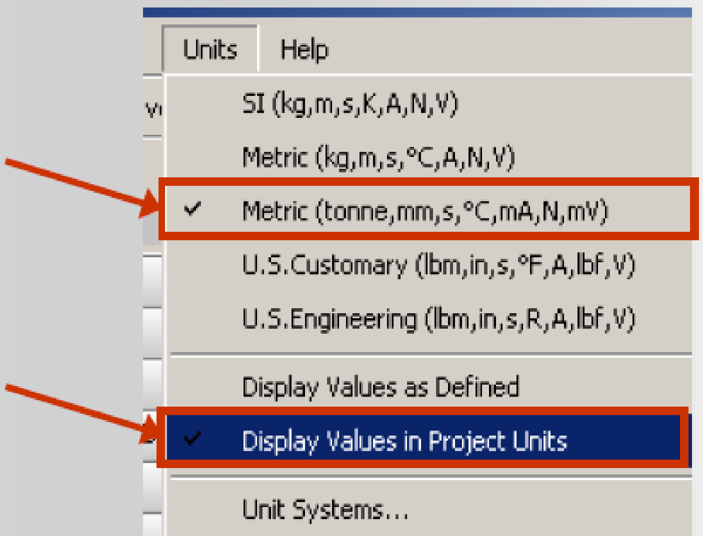


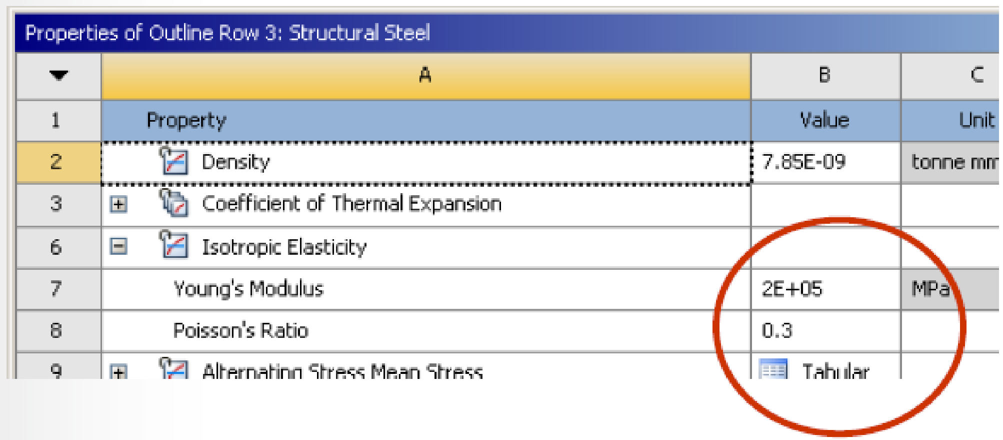
Note: The engineering data, geometry, loads and boundary conditions and preliminary contact region have already been set up. It remains to redefine the contact behavior and compare and contrast contact results.

… Workshop 3B: *Symmetric vs. Asymmetric*

Highlight the Engineering Data Cells double click to open and verify the predefined material properties

- **Verify that the units are in Metric(Tonne,mm,…) system. If not, fix this by clicking on…**
  - **Utility Menu>Units>Metric(Tonne, mm,..)**





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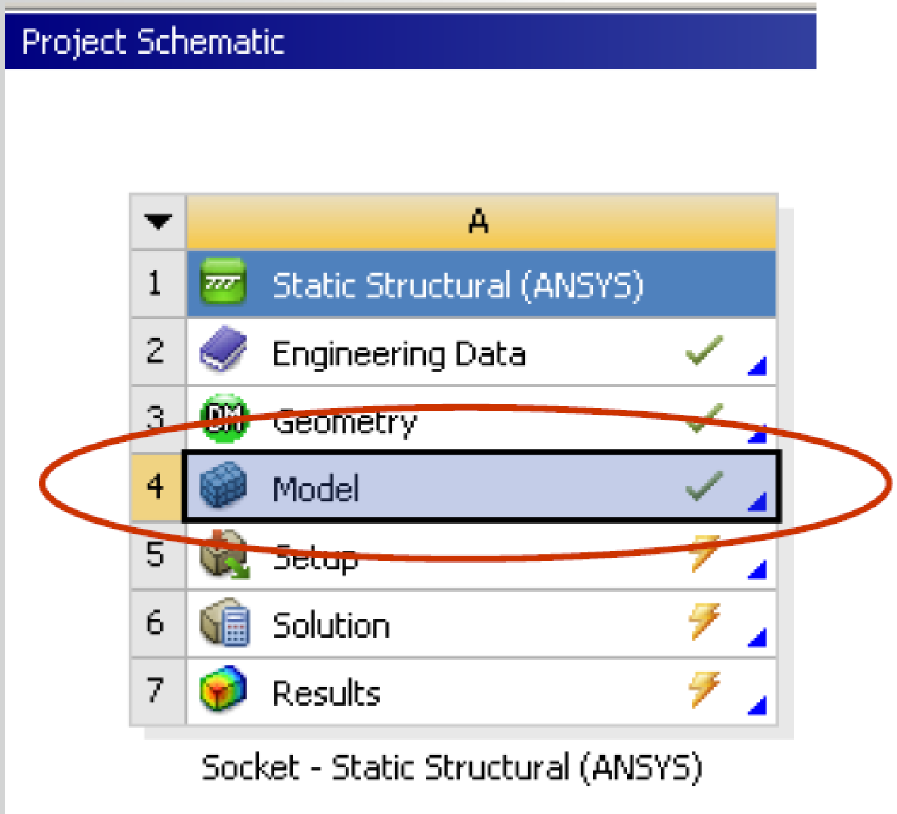
Return to the project schematic page

Double click (or RMB=>Edit…) on the Model Cell to open Mechanical Session



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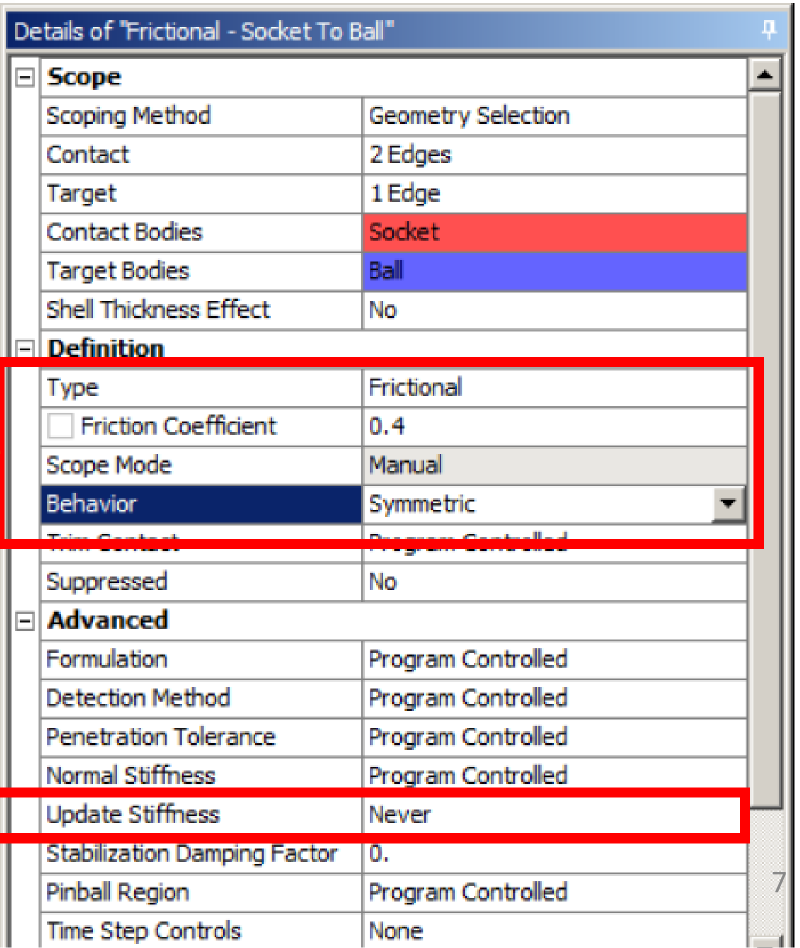
Once inside the Mechanical application, expand each folder in the project tree to become familiar with the model set up. Confirm material assignments, boundary conditions, and loads as described on the slide 2.

Highlight the contact region and modify the specifications as follows:

- **Type = Frictional**
- **Coefficient = 0.4**
- **Behavior = Symmetric**
- **Update Stiffness (Never)**



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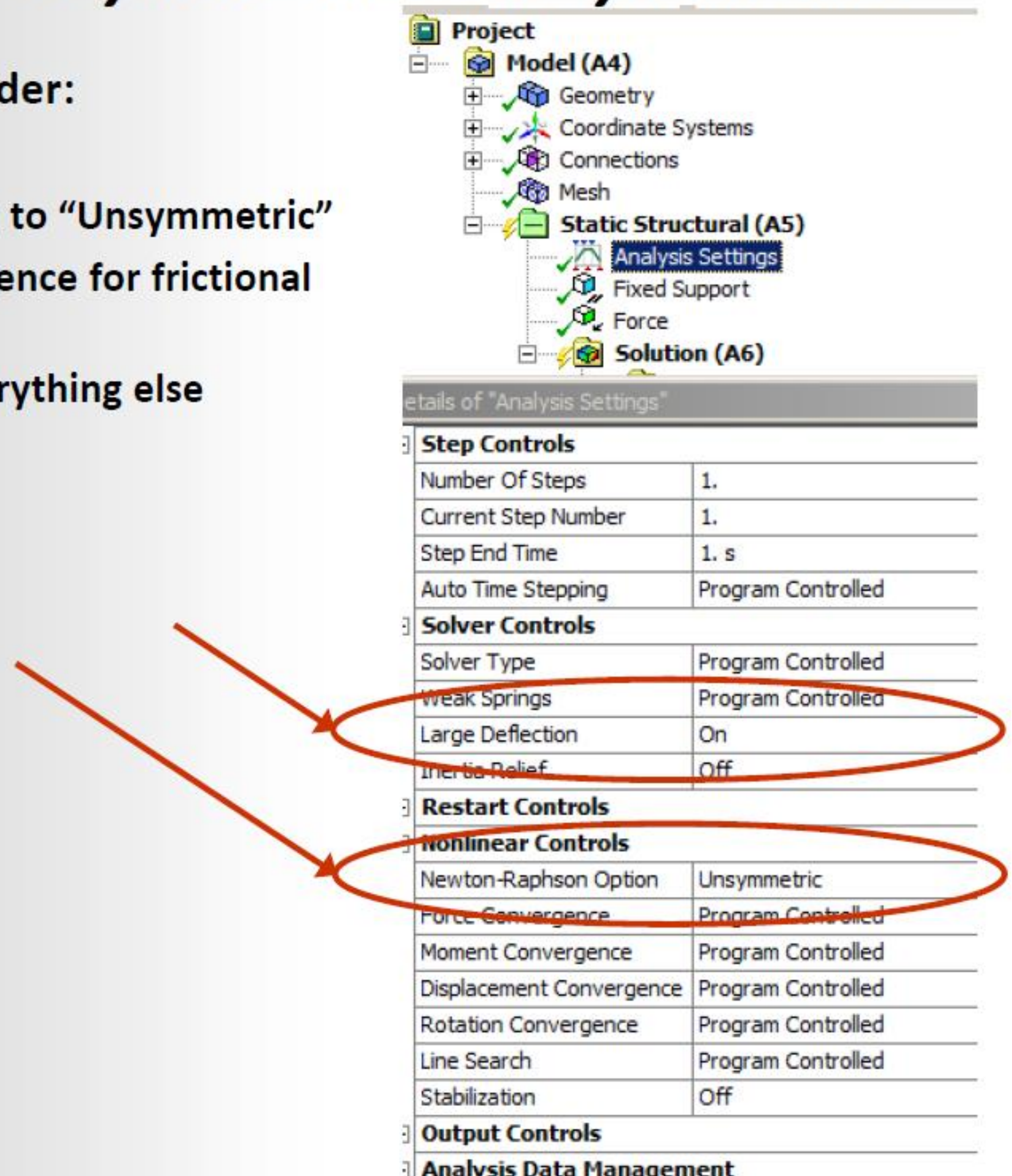
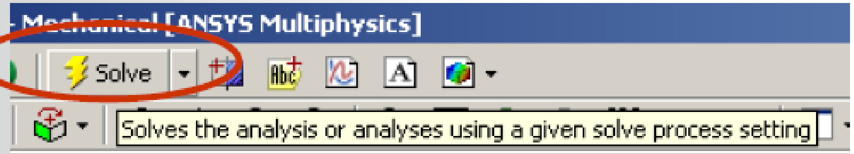
Highlight the Analysis Settings Folder:

- **Turn Large Deflection ON**
- **Set Newton-Raphson Option to “Unsymmetric”**
  - **This can enhance convergence for frictional contact problems.**
- **Take default settings for everything else**



... Workshop 3B: *Symmetric vs. Asymmetric*

Execute a Solve:

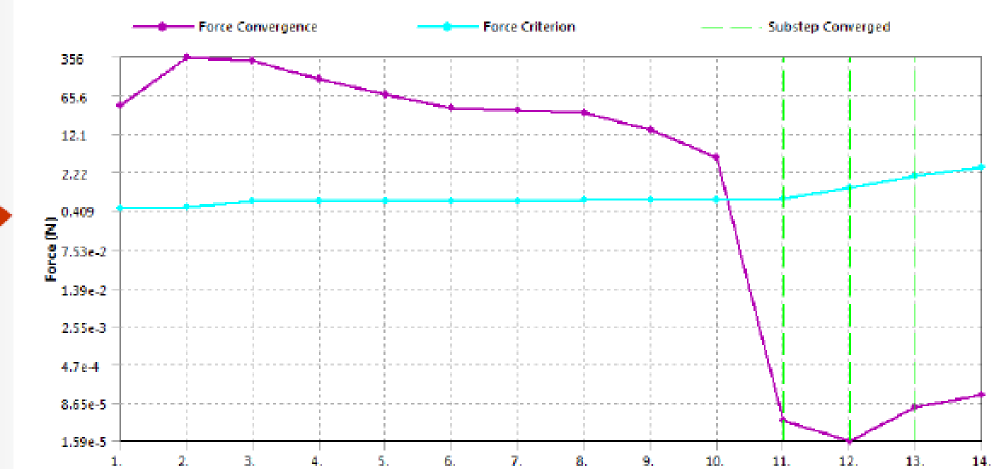



```
L O A D   S T E P   O P T I O N S

LOAD STEP NUMBER. . . . . . . . . . . . . . . . .     1
TIME AT END OF THE LOAD STEP. . . . . . . . . .  1.0000
AUTOMATIC TIME STEPPING . . . . . . . . . . . . .    ON
   INITIAL NUMBER OF SUBSTEPS . . . . . . . . .     5
   MAXIMUM NUMBER OF SUBSTEPS . . . . . . . . .    20
   MINIMUM NUMBER OF SUBSTEPS . . . . . . . . .     1
MAXIMUM NUMBER OF EQUILIBRIUM ITERATIONS. . . .    15
STEP CHANGE BOUNDARY CONDITIONS . . . . . . . .    NO
STEP-STIFFENING . . . . . . . . . . . . . . . .    ON
```

Highlight the Solution Information branch and note the following from the Solver output:

- **Because of the 0.4 friction coefficient, autotime stepping starts with 5 substeps.**
- **Turning on large deflection will ensure stress stiffening effects are included**
- **From the Force Convergence Graph, the solution converges easily**



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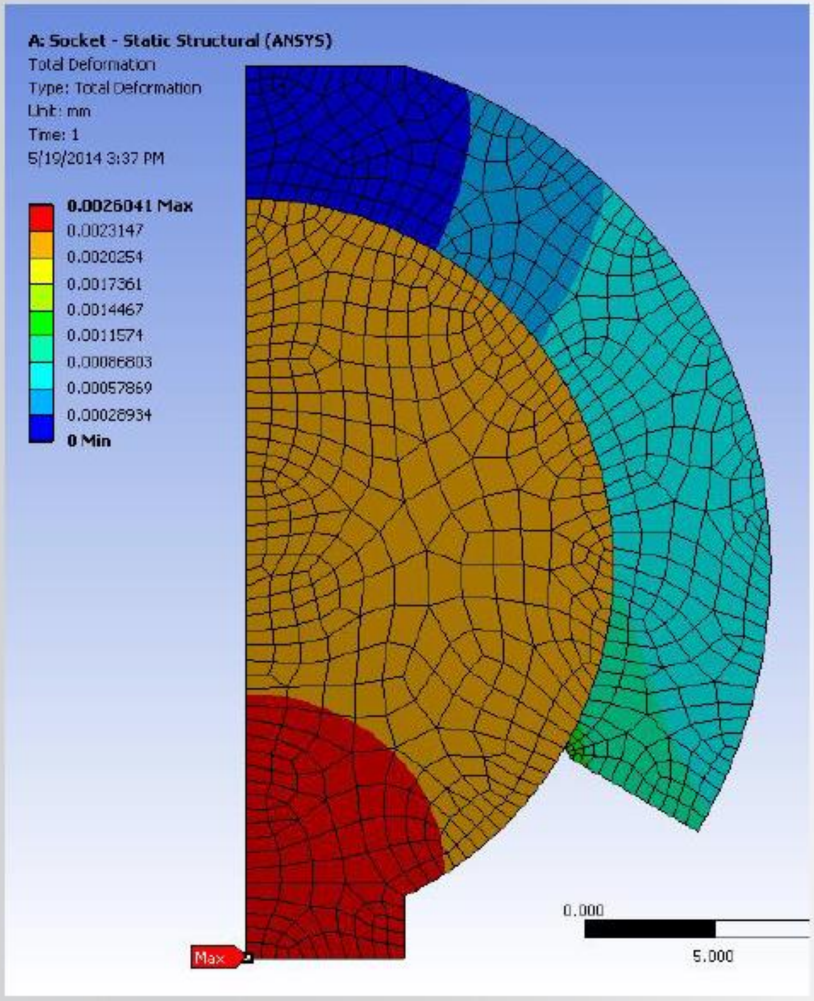
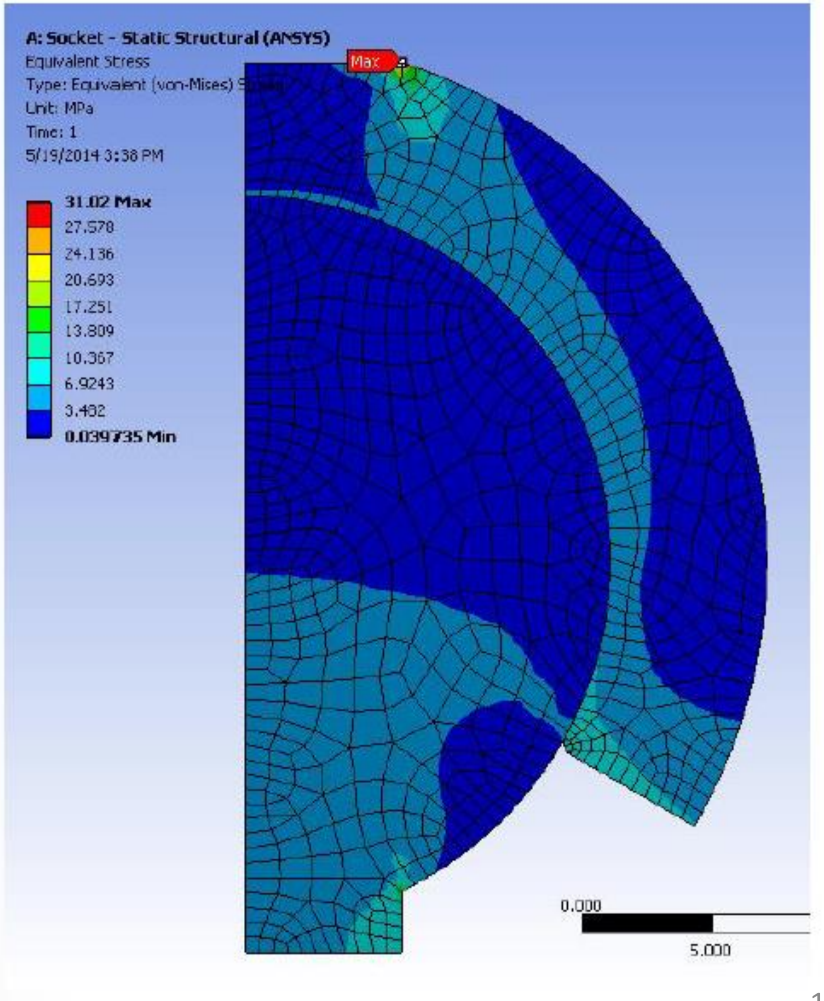
Post Process Total Deformation and Equivalent Stress

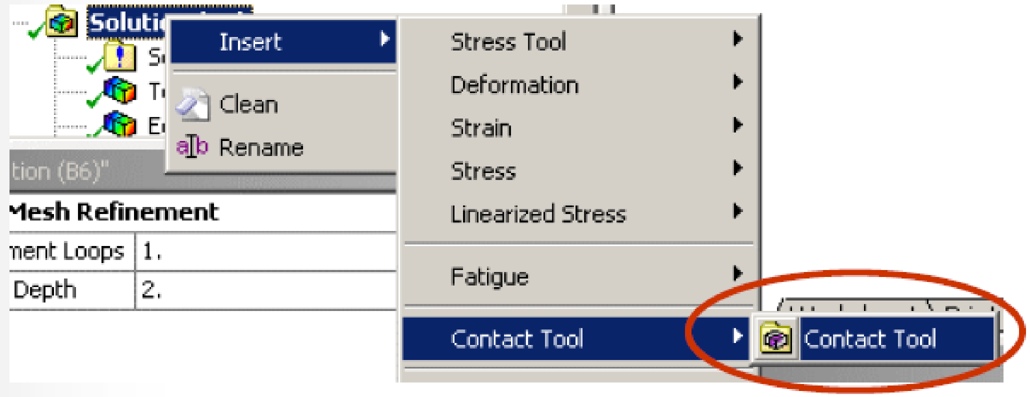




… Workshop 3B: *Symmetric vs. Asymmetric*

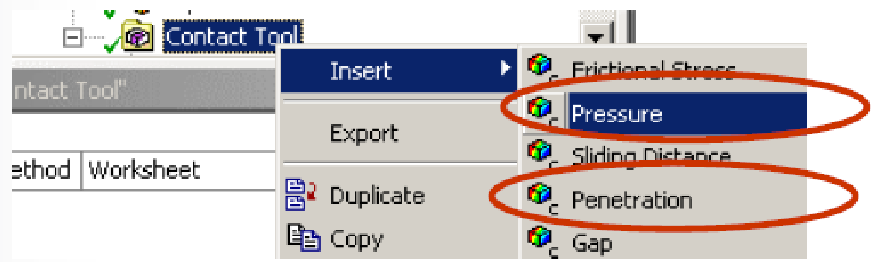
Highlight the Solution Branch

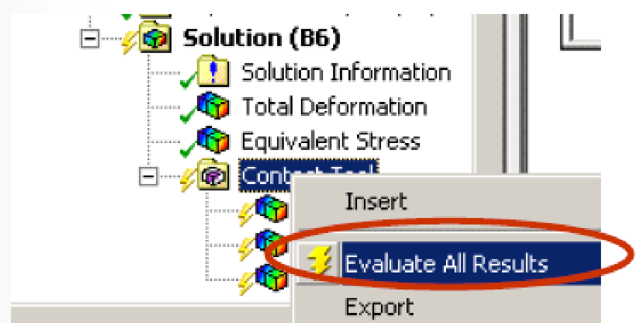
- **RMB > Insert > Contact Tool…**



Highlight the newly inserted Contact Tool

- **RMB > Insert >**
  - **Pressure**
  - **Penetration**
- **RMB > Evaluate Results**

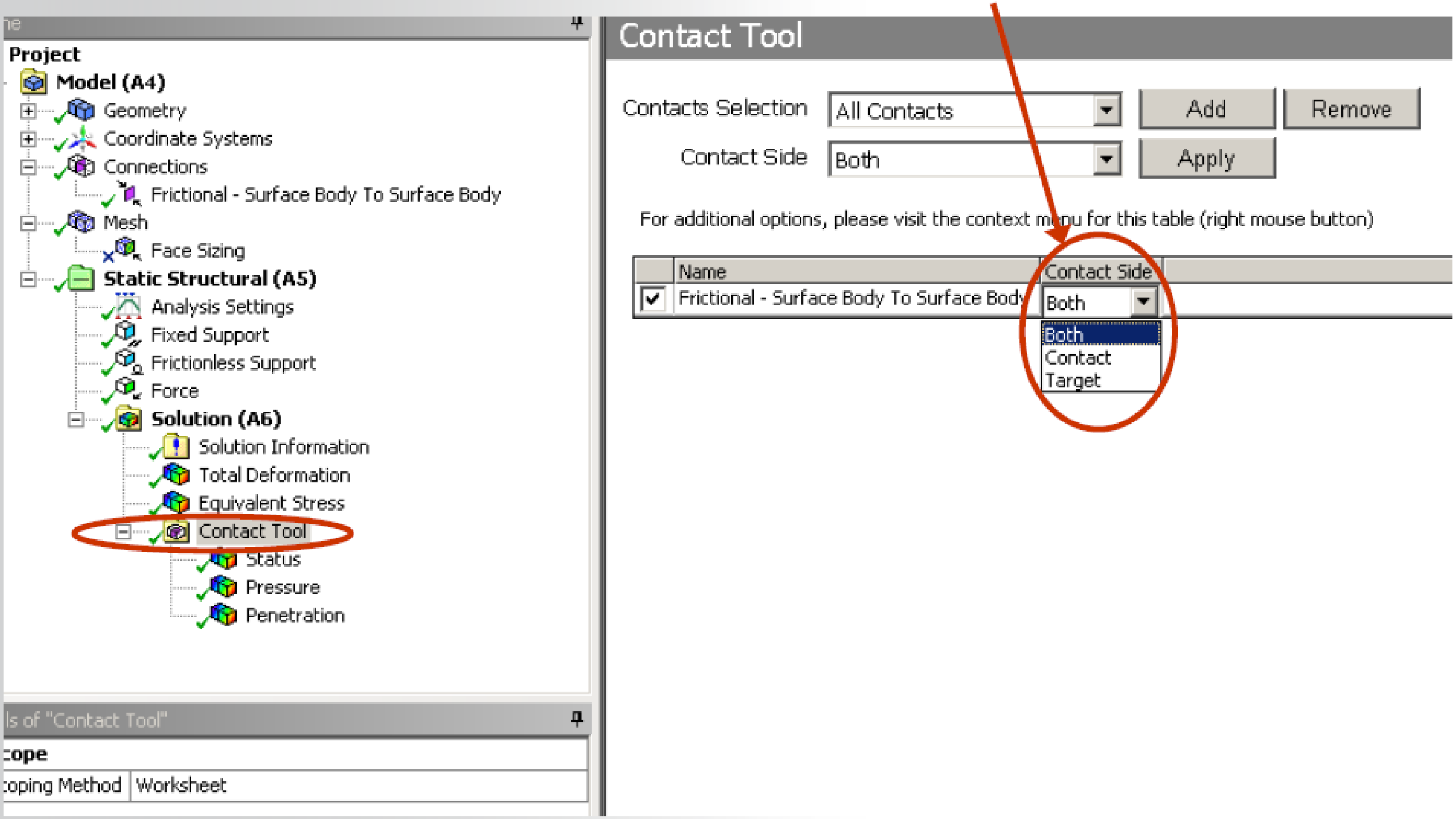




... Workshop 3B: *Symmetric vs. Asymmetric*

Recall, this is a symmetric contact region. Hence, there are contact results available on "both" the contact and target sides.

Insert 2nd Contact Tool, compare and contrast the results (status, pressure and penetration) for Contact Side = 'Contact' and Contact Side = 'Target'



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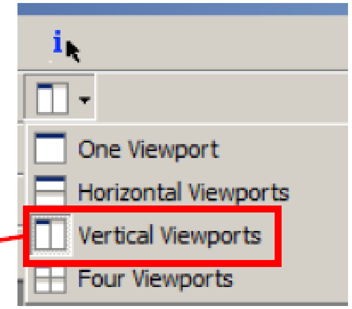
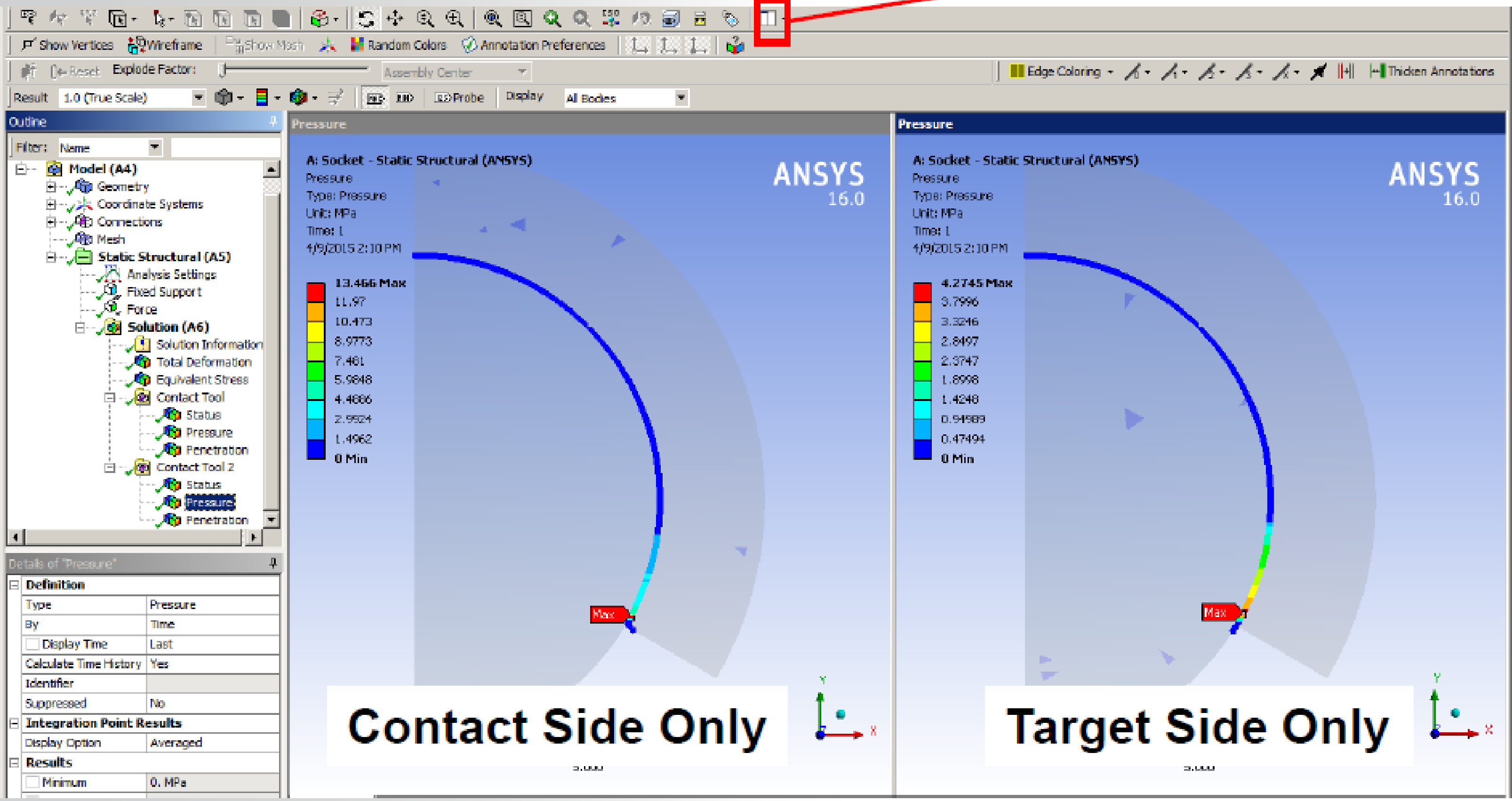


Use 'Vertical Viewports' option to view the two result sets side by side in Graphics Window.

Note the differences between the 'contact' and 'target' pressure profiles.

- Which one is 'correct'?



- Answer: Neither. The 'correct' answer is an average of the two profiles.

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Return to the Connection Folder.

Highlight the frictional contact region

- Change Behavior to Asymmetric

Rerun the solution

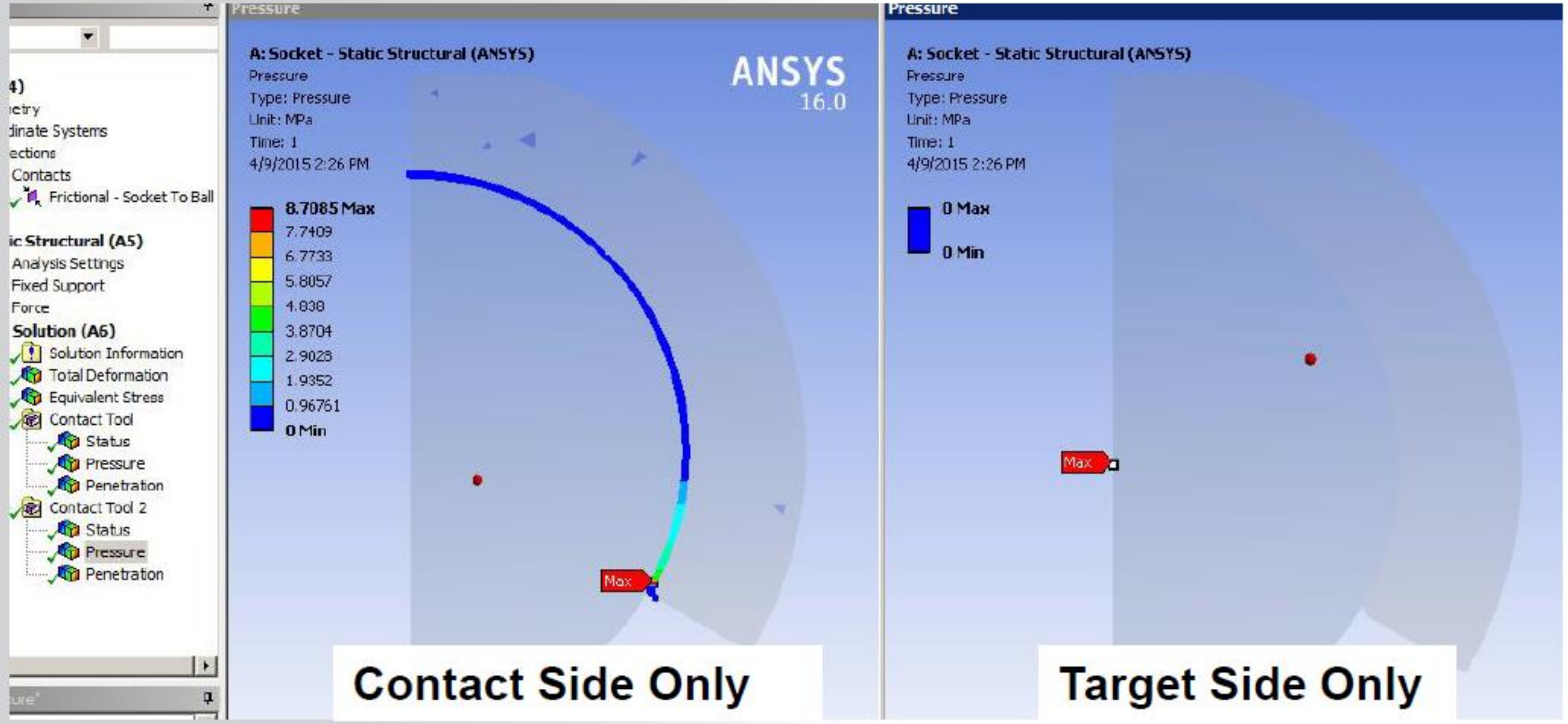
Details of "Frictional - Surface Body To Surface Body"

Scope	
Scoping Method	Geometry Selection
Contact	2 Edges
Target	1 Edge
Contact Bodies	Socket
Target Bodies	Ball
Definition	
Type	Frictional
Friction Coefficient	0.4
Scope Mode	Manual
Behavior	Asymmetric
Suppressed	No
Advanced	

... Workshop 3B: *Symmetric vs. Asymmetric*

Post process the contact results as before. Note that there is now only one answer (on the Contact side) and it is an approximate average of the two results that were available with the symmetric contact.

(13.466 + 4.2745)/2 = 8.87MPa*



***Note: Difference between hand calculation and FEA result is due to the assumption made in hand calc that max values are at exactly the same location at contacting interface. They are not in same location, but they are very close.**

… Workshop 3B: *Symmetric vs. Asymmetric*

Note that regardless of which contact behavior is used (Symmetric or Asymmetric) in this example, the overall model results for deformation and equivalent stress remain essentially the same. Symmetric behavior is intended to enhance convergence. However symmetric contact results can be more challenging to interpret.